11. LASSO — Predicting US Inflation

Removes weak predictors completely — when only some of the predictors are genuinely helpful.

LASSO Solves:

$$\hat{\beta}_{LASSO} = \arg\min_{\beta} \left[\sum_{i=1}^{n} (y_i - x_i'\beta)^2 + \lambda \underbrace{\sum_{j=1}^{p} |\beta_j|}_{\text{LASSO Penalty}} \right]$$

L_1 Penalty : Using absolute coefficients

LASSO can shrink coeff. exactly to zero.

LASSO performs automatic feature selection.

Sparse Model → A model where many coefficients are zero, so only a small # variables remain active.

The absol. value function has a sharp corner at zero — makes it mathematically easier for the optimisn. solution

to land exactly at $\beta_j = 0$.

LASSO Objective Function:

$$\hat{\beta}_{LASSO} = \arg\min_{\beta} \left[RSS + \lambda \sum_{j=1}^{p} |\beta_j| \right]$$

In Problem 12 (US Equity Premium) — LASSO eliminated all 14 predictors — became an intercept only model.

LASSO as a constraint problem:

$$\min_{\beta} RSS$$

subject to,

$$\sum_{j=1}^{p} |\beta_j| \leq c$$

$$\lambda^* = \arg\min_{\lambda} CV(\lambda)$$

Suppose if we had only two predictors: x_1, x_2

LASSO's constraint is:

$$|\beta_1| + |\beta_2| \leq c$$

This shape is a diamond. The diamond has sharp corners on the axes.

Those corners correspond to: $\beta_1 = 0$ or $\beta_2 = 0$

Because of these corners, the best solution often lands exactly on an axis. That means one coeff. becomes zero.

Soft Thresholding → Suppose one predictor and standardised data:

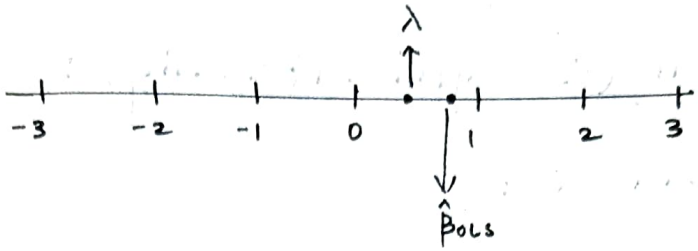
$$\hat{\beta}_{LASSO} = \begin{cases} \hat{\beta}_{OLS} - \lambda, & \hat{\beta}_{OLS} > \lambda \\ 0, & |\hat{\beta}_{OLS}| \leq \lambda \\ \hat{\beta}_{OLS} + \lambda, & \hat{\beta}_{OLS} < -\lambda \end{cases}$$

LASSO → Weak coeff. become zero
→ Strong coeff. become smaller

$\hat{\beta}_{OLS} > \lambda$

$(\hat{\beta}_{OLS} = 0.8)$

$(\lambda = 0.6)$



So, LASSO coeff. $(\hat{\beta}_{LASSO}) = \hat{\beta}_{OLS} - \lambda$

$= 0.20$

LASSO creates a dead zone around zero:

$$-\lambda \leq \hat{\beta}_{OLS} \leq \lambda$$

Any OLS coeff. that falls inside this zone is treated as too weak. So, $\hat{\beta}_{LASSO} = 0$

Hard Thresholding → Small coeff. ⇒ 0
→ Large coeff. ⇒ Unchanged

Soft Thresholding → Small coeff. ⇒ 0
→ Large coeff. ⇒ shrunk toward zero

$\lambda \uparrow \Rightarrow$ Wider Dead Zone $\Rightarrow$ More Coeff. become zero

Ordinary Cross Validn.: Randomly Split Data

Time series Cross Validn.: Uses Expanding or Rolling Windows

↳ Train on 1990 - 2000
↳ Validate on 2001
↳ Train on 1990 - 2001
↳ Validate on 2002 ..... so on.

In Problem 11: At the selected alpha, only the Federal Funds Rate remains.

Problem 11 (US Inflation Forecasting): Conceptually, the model is:

$$Inflation_{t+12} = \beta_0 + \beta_1 X_{1,t} + \beta_2 X_{2,t} + \cdots + \beta_p X_{p,t} + \varepsilon_{t+12}$$

The LASSO Version is:

$$\hat{\beta}_{Lasso} = \arg\min_{\beta} \left[\sum_t (Inflation_{t+12} - X_t'\beta)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right]$$

Our selected alpha was 0.2719

At this alpha:

↳ Fed. Funds Rate coeff. under LASSO = 0.5085

↳ " " " " under OLS = 1.2530

It is a predictive association and not causation.

LASSO can improve over unrestricted OLS:

In Problem 11:

$RMSE_{OLS} = 3.3681$ $\quad RMSE_{LASSO} = 2.5638$

$MAE_{OLS} = 2.1979$ $\quad MAE_{LASSO} = 1.6691$

Different metrics can favour different models.

RMSE punishes large errors more strongly.

MAE measures typical absolute error.

One major limitation: LASSO cannot automatically predict Structural Breaks.

Structural Break → A period when the old relationship b/w variables changes sharply.

E.g. In Problem 11, the post-pandemic inflation surge was a structural break.

A zero LASSO coefficient does not mean a variable is economically meaningless. It means the variable did not help enough in this predictive model, sample and validation design.

Large in-sample OLS coeff. do not guarantee ~~OBS~~ OOS predictive value.

Problem 12 (US Equity Premium Prediction) — LASSO became identical to the historical mean forecast. OLS performed worse. This means LASSO did not beat the benchmark but it protected against OLS overfitting.

Equity Premium R^2_{OOS} is:

$$R^2_{OOS} = 1 - \frac{\sum_t (y_t - \hat{y}_{model,t})^2}{\sum_t (y_t - \hat{y}_{benchmark,t})^2}$$

$R^2_{OOS} > 0 \Rightarrow$ model beats the benchmark

$R^2_{OOS} = 0 \Rightarrow$ Model matches the benchmark.

$R^2_{OOS} < 0 \Rightarrow$ Model worse than the benchmark.

Finance Lesson: In Equity Premium prediction, simple benchmarks are very hard to beat.

Directional accuracy can be misleading.

Problem 13 (GDP Growth & Climate Stress) — uses country-year panel data.

13 out of 14 predictors were retained by LASSO

A variable can be selected by LASSO and still fail to improve final test performance.

COVID was a tail event for GDP growth forecasting
↳ rare & extreme event.

If two variables are highly correlated, LASSO may arbitrarily keep one and drop the other — this motivates Elastic Net.

Common mathematical version of LASSO,

$$\hat{\beta} = \arg\min_{\beta} \left[\frac{1}{2n} \sum_{i=1}^{n} (y_i - x_i'\beta)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right]$$

For $|\beta|$ → Slope 1 from right side
→ Slope -1 from left side

Since $|\beta|$ is not differentiable at 0, we use a

subgradient

$$\partial |\beta| = \begin{cases} 1, & \beta > 0 \\ [-1, 1], & \beta = 0 \\ -1, & \beta < 0 \end{cases}$$

The range of possible slopes $[-1, 1]$, is what allows a coeff. to stay exactly at zero.

LASSO Objective:

$$L(\beta) = \frac{1}{2n} \| y - X\beta \|^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

First order optimality condition:

$$0 \in -\frac{1}{n} X'(y - X\beta) + \lambda \partial \|\beta\|_1$$

Residual is $r = y - X\beta$

$X'r$ measures how strongly each predictor is still correlated with leftover error.

For predictor j, the condition is

$$0 = -\frac{1}{n} x_j' (y - X\hat{\beta}) + \lambda s_j$$

$$\Rightarrow \quad \frac{1}{n} x_j' (y - X\hat{\beta}) = \lambda s_j$$

where $s_j \in \partial \|\hat{\beta}_j\|$

This gives us three cases (check subgradient $\partial|\beta|$)

Case 1: Positive coeff. $\hat{\beta}_j > 0$

then $s_j = 1$

So, $\frac{1}{n} x_j' r = \lambda$

Case 2: -ve coeff. $\hat{\beta}_j < 0$

Then $s_j = -1$

So, $\frac{1}{n} x_j' r = -\lambda$

Case 3: zero coeff.

$\hat{\beta}_j = 0$, then $s_j \in [-1, 1]$

so, $\frac{1}{n} x_j' r \in [-\lambda, \lambda]$

$\Rightarrow \hat{\beta}_j = 0$ when $\left|\frac{1}{n} x_j' r\right| \leq \lambda$ (KKT Zero Condition)

i.e., if a predictor is not sufficiently correlated with the remaining unexplained part of y, LASSO sets its coeff. to zero.

If $\left|\frac{1}{n} x_j^T r\right| > \lambda$, then the predictor is strong enough to survive.

λ acts like a gatekeeper.

Exact Soft-Thresholding Formula:

In the simplest case, where predictors are standardised and orthogonal:

$$S(z, \lambda) = \text{sign}(z)\,(|z| - \lambda)_+$$

where, $(a)_+ = \max(a, 0)$

This means:

$$S(z, \lambda) = \begin{cases} z - \lambda, & z > \lambda \\ 0, & |z| \le \lambda \\ z + \lambda, & z < -\lambda \end{cases}$$

So, the LASSO coeff. becomes:

$$\hat{\beta}_j^{\text{LASSO}} = S\left(\hat{\beta}_j^{\text{OLS}}, \lambda\right)$$

under the simple orthogonal design case

Ridge has a closed form solution.

LASSO → $|\beta_j|$ has non-differentiability issues at zero.

So, LASSO is usually solved using iterative algorithms

— e.g. Coordinate descent

↳ Update one coeff. at a time while holding the others fixed.

For one coeff. j, the update looks like:

$$\beta_j \leftarrow \frac{S\left(\frac{1}{n} x_j' r_j, \lambda\right)}{\frac{1}{n} x_j' x_j}$$

where,

$$r_j = y - \sum_{k \neq j} x_k \beta_k$$

↳ Residual after removing the effect of all predictors except x_j

λ_{max}: smallest value of λ that makes all slope coefficients zero.

Mathematically, assuming y is centred and predict are standardised

$$\lambda_{max} = \max_j \left| \frac{1}{n} x_j' y \right|$$

If $\lambda \geq \lambda_{max}$, then:

$$\hat{\beta}_1 = \hat{\beta}_2 = \ldots = \hat{\beta}_p = 0$$

Model becomes intercept only.

↳ Look at how strongly each predictor is correlated with the target. The largest absolute correln. determines the penalty needed to kill all predictors.

The efficient degrees of freedom of LASSO are roughly the # selected predictors:

$$df_{LASSO} \approx \#\{j : \hat{\beta}_j \neq 0\}$$

LASSO coefficients should be interpreted cautiously.

Mathematically, when predictors are correlated, the LASSO objective can have several very similar solutions in terms of prediction error, but different selected variables. This is one reason Elastic Net exists.

Selection Consistency → As the sample size becomes very large, does LASSO correctly identify the true set of relevant variables?

If irrelevant variables are highly correlated with relevant variables, LASSO may select the wrong one.

Post-LASSO OLS → Use LASSO to select variables
Refit OLS using only the selected variables.

But this also has risks. If LASSO selected the wrong variables, POST-LASSO OLS may still be flawed.

Also, normal OLS p-values after selection are not straightforward because the selection step changes the statistical problem.

LASSO is not designed for classical inference.

In OLS, we often do:

$$t = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}$$

and then calculate p-values.

With LASSO this is harder, because the model selection and coefficient estimation happen together. The same data are used to decide — which variables enter and what coeff. they get. This means ordinary p-va after LASSO are not automatically valid.

LASSO is mainly a prediction and variable-selection metho Classical inference after LASSO requires special care.

As λ changes, the LASSO coeff. move along a piecewise linear path.

As λ decreases from a very large value:

- ↳ all coeff. start at zero.
- ↳ strongest predictor enters first
- ↳ more predictors enter as penalty weakens
- ↳ coeff. grow or change direction along the path